EuroSkills 2027 – IT Network Systems Administration

National Semi-final

Module B

28th February 2026

Grüß Gott!

We would like to thank you for the work you carried out on the Paulaner network. Your contribution was extremely helpful.

As you did an excellent job, we now require your assistance while preparing for Oktoberfest. This time, however, the task is not related to the network itself but rather to preparing the servers to ensure everything runs smoothly.

The servers already have been pre-configured, and you can find further information about the infrastructure below.

Information about the infrastructure

You can access the VMware infrastructure at **https://vcsa01.s39.lan**

In the following table (*Table 1*), you can find details about the addressing of the devices, their operating systems, and the pre-installed packages.

Name	IP	OS	Pre-installed packages
OKT-DC	2001:db8:f145:10::10/64	Windows Server 2022 (Datacenter)	Active Directory Domain Services, DNS, SSH server
OKT-FILE	2001:db8:f145:10::11/64	Windows Server 2022 (Datacenter)	SSH server
OKT-BACKUP	2001:db8:f145:10::20/64	Debian 12.10	syslog-ng, nftables
OKT-WEB	2001:db8:f145:10::21/64	Debian 12.10	nginx, syslog-ng, nftables, openssl
OKT-CLT	DHCP	Windows 11	Mozilla Firefox, Putty, WinSCP

Table 1 – Information about the computers

Users

Name	Username	Groups	Department
Karla Baumann	karla.bauma	Waiter, Employee	Tents
Ludwig Schlimme	ludwig.schli	TicketHandler, Employee	Selling
Kurt Klein	kurt.klein	Manager	HQ
Gunter Bergmann	gunter.bergm	IT, Employee	IT

Table 2 – Users

Topology

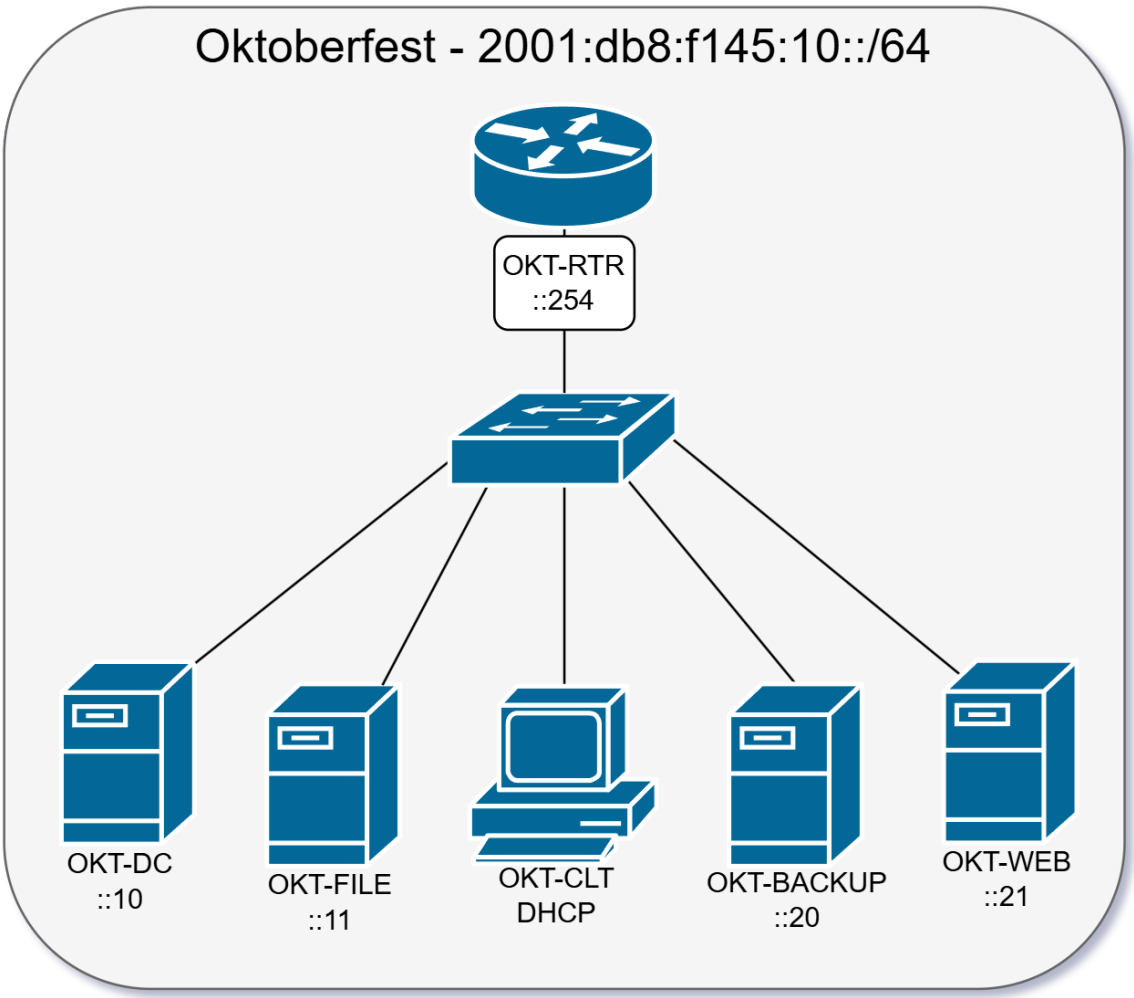

Figure 1 – Logical topology

Additional Informations

- All computers have EN/US keyboard layout configured!
- Unless otherwise specified, use the password: “**1MassB!tt3**”!
- Our domain name is **paulaner.de**
- The address of the gateway is **2001:db8:f145:10::254**
    - The router is configured to send IPv6 Neighbor Discovery RA messages with 'managed' and 'other' flags. It has already been preconfigured, and you don't have access to it.
- The usage of snapshots is allowed, so use them as required!
- You must assign every IPv6 address on the network!
    - If you can’t make the services work with IPv6 you can use IPv4 for significantly less points.
- You must assign to every device its hostname!
- Pre-installed packages for testing purposes on Linux machines:
    - curl, dnsutils, wget, telnet, traceroute, net-tools, nmap, tree, openssl, ssh
- Use the username Admin to log on to OKT-CLT (without AD)!
- Test every change you make!
- At the end of the allocated time, leave every server as they are!

Tasks

OKT-DC

1. Create the following OUs using **PowerShell** and place the commands you have issued to **"C:\Scripts\ou.ps1"** file with the exact path and name. (If you can't create them with PowerShell, you can create them manual by GUI for less points.) The OU structure is in *Table 3*.

OU Path	Description
Oktoberfest	-
Oktoberfest\Groups	Group objects
Oktoberfest\Users	-
Oktoberfest\Users\HQ	Users from the HQ
Oktoberfest\Users\IT	IT Department users
Oktoberfest\Users\Tents	Waiters, beer servers
Oktoberfest\Users\Selling	Ticket management

Table 3 – OU structure

2. Set up the groups and users in their appropriate locations and properties, making sure that everyone belongs to the correct group as shown in Table 2. Except for managers, all users should be part of the Employee group.
3. You have to **create folders using** a **PowerShell** script. Place the commands you issued to **"C:\Scripts\folders.ps1"**. (If you can't create them with PowerShell, you can create them by hand for less points.) After the folders are created, give the permissions listed below to ensure proper access both locally (NTFS) and remotely (SMB)!
   a. C:\Paulaner - Make sure you can use it with DFS!
   b. C:\Paulaner\Beers - "**Waiter"** group members can read it, and **Managers** have full access.
   c. C:\Paulaner\Tickets - "**TicketHandler"** group members can read it, and "**Manager"** group members have full access.
   d. C:\Paulaner\Employees - All users except Managers have write and read permissions.
   e. C:\Paulaner\Manager - Only "**Manager"** group members can read and write in this folder.
   f. IT Users must have permission to manage all these folders!
4. Install and set up **DFS** on OKT-DC. Replicate all folders above to OKT-FILE using it. Use the name of your choice when creating the namespace and replication group!
5. Create a new **DHCPv6** Scope using the following settings:

   a. Clients get address from 2001:db8:f145:10::/64 subnet, starting from 2001:db8:f145:10:1::0!
   b. Configure OKT-DC as a DNS server!
   c. Add domain name (paulaner.de)!
   d. Make sure every DHCP client gets a DNS record!

6. Create appropriate records for all servers based on their hostnames. Create these CNAME records as well:
   a. www - OKT-WEB.paulaner.de
   b. intra - OKT-WEB.paulaner.de
   c. syslog - OKT-BACKUP.paulaner.de

7. Create the following **GPOs**
   a. Make every domain joined computer to get the DFS namespace root automatically mounted!
   b. Deploy the Root CA created on OKT-WEB!

OKT-FILE

1. Join the computer to **paulaner.de** domain!
2. Create the folders with sufficient permission as on OKT-DC (task 3).
3. Install **DFS** and make sure folders are replicated to this computer from OKT-DC.

OKT-BACKUP

1. Create a **nftables** rule that creates a log message for every **incoming** packet, that is destined to this computer.
2. Set up **Syslog-NG server** for receiving logs
   a. Logs from OKT-WEB.paulaner.de
      i. Place access logs for *www.paulaner.de* to ***/var/ng-logs/www.log***
      ii. Place access logs for *intra.paulaner.de* to ***/var/ng-logs/intra.log***
3. Create a backup from your syslog-ng config file every 20 minutes to /backup/$DATE/syslog.conf, $DATE represents a folder with the current date in the following format: '**+%Y-%m-%d_%H:%M**'!

OKT-WEB

1. Create a **nftables** rule that creates a log message for every **incoming** packet, that is destined to this computer.
2. Create a Root CA for your company with the following attributes:
   a. Place the file to /ca/CA.crt
   b. CN=Oktoberfest CA
   c. O=Paulaner

   d. C=DE
   e. 2 years
3. Generate a server certificate signed by your Root CA that includes:
   a. The server's FQDN as the Subject Common Name (CN)
   b. All DNS names and IPv6 addresses of the OKT-WEB server in the Subject Alternative Name (SAN) extension
   c. Place all files under /ca directory
4. Configure **nginx**:
   a. Create a virtual host that serves the site under /var/www/html
      i. Default index must be iisstart.htm.
      ii. It must serve **www.paulaner.de**.
      iii. It must listen only on the HTTPS port using the certificate you have created.
   b. Create a virtual host that serves the site under **/var/www/intra**
      i. Default index must be paulaner.html.
      ii. It must serve **intra.paulaner.de**.
      iii. It must listen only on the HTTPS port using the certificate you have created.

OKT-CLT

1. Connect the client to the domain and login with Kurt Klein and Karla Baumann Active Directory account!
2. Test the connection to the webserver!
3. Make sure you test everything, and it's working well!

Congratulations, you have reached the end of this part of the project!